CM0845 Logic
First-Order Logic: Syntax

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Preliminaries

Textbook

Dirk van Dalen (2013). Logic and Structure. Sections 3.1, 3.2 and 3.3.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Introduction

Example (informal examples (p. 53))

$\exists x P(x)$	there is an x with property P
$\forall y P(y)$	for all y P holds (all y have the property P)
$\forall x \exists y (x = 2y)$	for all x there is a y such that x is two times y
$\forall \epsilon (\epsilon > 0 \rightarrow \exists n (n < \epsilon))$	for all positive there is an n such that $n < \epsilon$
$x < y \rightarrow \exists z (x < z \wedge z < y)$	if $x < y$, then there is a z such that $x < z$ and $z < y$
$\forall x \exists y (x \cdot y = 1)$	for each x there exists an inverse y

Structures

Definition 3.2.1

A **structure** is an ordered sequence

$$\langle A, R_1, \ldots, R_n, F_1, \ldots, F_m, \{c_i \mid i \in I\}\rangle,$$

where

(i) A is a non-empty set, the **universe** of the structure,

(ii) $R_1, \ldots, R_n$ are relations on A,

(iii) $F_1, \ldots, F_m$ are functions on A, and

(iv) the c_i, where $i \in I$, are elements of A (constants).

Notation

- Structures are denoted by Gothic capitals: $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \ldots$
- $A = |\mathfrak{A}|$

Structures

Examples

Whiteboard.

Structures

Definition 3.2.2

The **similarity type** (or **signature** or **non-logical constants**) of a structure

$$\langle A, R_1, \ldots, R_n, F_1, \ldots, F_m, \{c_i \mid i \in I\}\rangle$$

is a sequence

$$\langle r_1, \ldots, r_n; a_1, \ldots, a_m; \kappa\rangle,$$

where

(i) $R_i \subseteq A^{r_i}$,

(ii) $F_j : A^{a_j} \to A$, and

(iii) $\kappa = |\{c_i \mid i \in I\}|$ (cardinality of I).

Structures

Examples

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Structures

Examples

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Limiting cases

0-ary relations and 0-ary functions.

Structures

Examples

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Limiting cases

0-ary relations and 0-ary functions.

Convention

All the structures are equipped implicitly with the identity relation.

Alphabet

Definition (p. 56)

The **alphabet** has the following symbols:

(i) Predicate symbols: $P_1, \dots, P_n$ and $\doteq$

(ii) Function symbols: $f_1, \dots, f_m$

(iii) Constant symbols: $\overline{c}_i$ for $i \in I$

(iv) Variables: $x_0, x_1, x_2, \dots$ (countably many)

(v) Connectives: $\vee, \wedge, \rightarrow, \neg, \leftrightarrow, \bot, \forall, \exists$

(vi) Auxiliary symbols: $(,)$

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(v) Connectives: $\vee, \wedge, \rightarrow, \neg, \leftrightarrow, \perp, \forall, \exists$

(vi) Auxiliary symbols: $($, $)$

Remark

The equality symbol.

The Set of Terms

Definition 3.3.1

The **set of terms**, denoted TERM, is the smallest set X with the properties:

(i) $x_i \in X$, where $i \in \mathbb{N}$,

(ii) $\overline{c}_i \in X$, where $i \in I$, and

(iii) $t_1, \ldots, t_{a_i} \in X \Rightarrow f_i(t_1, \ldots, t_{a_i}) \in X$, for $1 \leq i \leq m$.

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Examples

Whiteboard.

The Set of Formulae

Definition 3.3.2

The **set of formulae**, denoted FORM, is the smallest set X with the properties:

(i) $\perp \in X$,

(ii) $P_i \in X$ if $r_i = 0$,

(iii) $t_1, \ldots, t_{r_i} \in \mathrm{TERM} \Rightarrow P_i(t_1, \ldots, t_{r_i}) \in X$,

(iv) $t_1, t_2 \in \mathrm{TERM} \Rightarrow t_1 \doteq t_2 \in X$,

(v) $\varphi, \psi \in X \Rightarrow (\varphi \, \square \, \psi) \in X$, where $\square \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$,

(vi) $\varphi \in X \Rightarrow (\neg\varphi) \in X$,

(vii) $\varphi \in X \Rightarrow ((\forall x_i)\varphi) \in X$ and

(viii) $\varphi \in X \Rightarrow ((\exists x_i)\varphi) \in X$.

The formulae defined in the four first items are called atomic formulae or atoms.

Notational Conventions

- We use the conventions of propositional logic.
- We delete the outer brackets and the brackets round $\forall x$ and $\exists x$ whenever possible.
- Quantifiers bind more strongly than binary connectives.
- Join strings of quantifiers, e.g. $\forall x_1 x_2 \exists x_3 x_4 \varphi$ stands for $\forall x_1 \forall x_2 \exists x_3 \exists x_4 \varphi$.

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Examples

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Induction Principles and Recursive Definitions

Remark

Given that TERM and FORM are set inductively defined, we have induction principles and recursive definitions on them.

Set of Free Variables of a Term

Definition 3.3.6

The **set of free variables of a term** t, denoted $\mathrm{FV}(t)$, is defined by

$$\begin{aligned}
\mathrm{FV} : \mathrm{TERM} \to \{x_i \mid i \in \mathbb{N}\}& \\
\mathrm{FV}(x_i) &= \{x_i\}, \\
\mathrm{FV}(\bar{c}_i) &= \emptyset, \\
\mathrm{FV}(f(t_1, \ldots, t_n)) &= \mathrm{FV}(t_1) \cup \cdots \cup \mathrm{FV}(t_n).
\end{aligned}$$

Closed Terms

Definition 3.3.8

A term t is **closed** iff $\mathrm{FV}(t) = \emptyset$. The set of closed terms is denoted by TERM_c.

Examples

Whiteboard.

Set of Free Variables of a Formula

Definition 3.3.7

The **set of free variables of a formula** φ, denoted $\mathrm{FV}(\varphi)$, is defined by

$$
\begin{aligned}
\mathrm{FV} &: \mathrm{FORM} \to \{x_i \mid i \in \mathbb{N}\} \\
\mathrm{FV}(\bot) &= \emptyset \\
\mathrm{FV}(P) &= \emptyset, \text{ for } P \text{ propositional symbol} \\
\mathrm{FV}(P(t_1, \ldots, t_n)) &= \mathrm{FV}(t_1) \cup \cdots \cup \mathrm{FV}(t_n), \\
\mathrm{FV}(t_1 \doteq t_2) &= \mathrm{FV}(t_1) \cup \mathrm{FV}(t_2), \\
\mathrm{FV}(\varphi \square \psi) &= \mathrm{FV}(\varphi) \cup \mathrm{FV}(\psi), \\
\mathrm{FV}(\neg\varphi) &= \mathrm{FV}(\varphi), \\
\mathrm{FV}(\forall x_i \varphi) = \mathrm{FV}(\exists x_i \varphi) &= \mathrm{FV}(\varphi) - \{x_i\}.
\end{aligned}
$$

Sentences

Definition 3.3.8

A formula φ is **closed** iff $\mathrm{FV}(\varphi) = \emptyset$. A closed formula is also called a **sentence**. The set of sentences is denoted by SENT.

Examples

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Definition 3.3.12

A **term** t **is free for a variable** x **in a formula** φ iff

(i) φ is atomic,

(ii) $\varphi := \neg\psi$ and t is free for x in ψ,

(iii) $\varphi := \varphi_1 \square \varphi_2$ and t is free for x in φ_1 and φ_2,

(iv) $\varphi := \forall y \psi$ and if $x \in \mathrm{FV}(\varphi)$, then $y \notin \mathrm{FV}(t)$ and t is free for x in ψ, or

(v) $\varphi := \exists y \psi$ and if $x \in \mathrm{FV}(\varphi)$, then $y \notin \mathrm{FV}(t)$ and t is free for x in ψ.

The Extended Language

Definition 3.3.12

The **extended language**, $L(\mathfrak{A})$, of $\mathfrak{A}$ is obtained from the language L, of the type of $\mathfrak{A}$, by adding constant symbols for all elements of $|\mathfrak{A}|$. We denote the constant symbol, belonging to $a \in |\mathfrak{A}|$, by $\overline{a}$.

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Examples

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References

Dirk van Dalen [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on p. 2).